CS203 Midterm — Student Answer Sheet

Week 8 review | Version A | Questions 1–26 | For learning and self-checking

Part 1 — Multiple-choice key and brief reasoning (Q1–Q20)

Q1 — D. The target is rental duration across rentals, so each completed rental contributes one observation. A bike-level mean would answer a different question.

Q2 — C. Broader planned coverage addresses exclusions of other stations, times and days. Merely enlarging the same restricted sample does not remove that coverage problem.

Q3 — A. The numerator is the number of distinct bikes used, not the number of trips. One of the four fleet bikes was used, so the fraction is 1/4.

Q4 — A. The middle two sorted values remain 5 and 5. The mean rises from 20/4 = 5 to 92/4 = 23 minutes; an unchanged median does not imply unchanged data.

Q5 — C. Multiplying all values by two multiplies the mean by two and every squared deviation, hence the variance, by four.

Q6 — D. For A the squared deviations sum to 200, giving sample variance 100. For B they sum to 2, giving variance 1. Equal means do not imply equal variability.

Q7 — D. The displayed formula divides by zero. A software convention for a constant feature must be specified explicitly; it is not supplied by the z-score formula.

Q8 — B. The records conflict with the declared key. Different values do not establish which record is correct; provenance or source verification is needed before correction.

Q9 — C. A valid unit conversion gives 2 × 60 = 120 minutes. The large resulting value is not by itself a reason to discard a source-confirmed observation.

Q10 — B. Not checked is an unknown result, whereas zero is an observed negative result. Conflating them adds unsupported healthy outcomes to the data.

Q11 — D. Cleaning the source-confirmed extra copy leaves two genuine observed counts, 0 and 100, whose median is 50. Fitting before removing the copy weights the duplicated observation twice.

Q12 — B. The valid training values are 2, 4 and 6, whose median is 4. The held-out value must not influence the fitted imputation statistic.

Q13 — A. Only the earlier-rental count is stated to be available by 12:00. Features created at return or repair use information from after the prediction cutoff.

Q14 — B. The model needs the same fitted representation at training and evaluation. Applying the fixed training bounds gives 40/100 = 0.4 in either set; separately fitting on test data changes that representation.

Q15 — A. The supplied pairs form a U-shaped pattern: zero linear Pearson correlation does not exclude a nonlinear relationship. No Pearson formula or invariance theorem is required.

Q16 — A. Weather may relate to both variables and produce or alter their pooled association. Within-weather comparisons help inspect that explanation but do not alone establish causality.

Q17 — C. Correlation can be sensitive to an influential point. The old bike may have low usage, incomplete records or a valid unusual history; the changed statistic alone does not identify an error.

Q18 — D. A paired scatter plot displays both numerical measurements for each bike and can show linear or curved patterns. Separate summaries lose those links.

Q19 — B. A changes from 1/2=50% to 2/2=100%, a 50-percentage-point increase. B changes from 20/40=50% to 21/40=52.5%, an increase of 2.5 points. A smaller denominator makes a single changed outcome more influential.

Q20 — C. 120/6=60/3=20 minutes per trip, but the underlying totals differ. A ratio-only feature discards information about the total activity; preserve totals/counts when the task needs them.

Part 2 — Worked written answers (Q21–Q26)

Q21 — Counting bikes in service records

(a) The observation unit is one distinct fleet bike. B17 and B24 had at least one service job; B31 had none. The serviced share is 2/3 = 66.67%.

(b) An inner join removes B31 (no job) and produces two rows for B17 (jobs J1 and J2). The three matching job rows therefore represent only two bikes. Start from all three fleet bikes and left-join a serviced flag, or count distinct serviced bike IDs divided by all fleet bike IDs. Count B17 once and B31 as unserviced. An unaggregated left join has four job/bike rows and still requires bike-level aggregation.

(c) The weekday-before-09:00 East-station survey may miss West-station riders, afternoon/evening riders, and weekend riders. Sample across stations, times of day, and weekdays/weekends with a documented sampling method.

Q22 — Comparing two repair records

(a) East values: 2, 2, 4, 4, 8. Mean = (2+2+4+4+8)/5 = 4 minutes; median = 4 minutes; range = 8 − 2 = 6 minutes.

(b) The same mean and median do not imply identical variability or distribution. East ranges from 2 to 8 (range 6); West ranges from 3 to 5 (range 2). West is more consistent in the supplied records. These observations cannot establish identical or different future behaviour.

(c) Adding a valid East value of 0 lowers the mean, since 0 is below the old mean: new mean = 20/6 ≈ 3.33. The minimum drops to 0, maximum stays 8, and range increases to 8 − 0 = 8 minutes.

Q23 — Cleaning rental data with different units

(a) Keep one T02 record and remove the confirmed duplicate. Convert 0.5 hour to 30 minutes while retaining units and provenance. Flag −4 minutes as invalid/unknown, not 0 or +4, and ask for a source correction. Retain the confirmed genuine 240-minute rental despite its large value.

(b) Only four of five genuine rentals have valid observed durations: 12, 30, 15, 240. Their mean is (12+30+15+240)/4 = 74.25 minutes. This is the mean of four observed valid durations, not the known mean for all five rentals; the missing value could alter it.

(c) Sorted observed durations are 12, 15, 30, 240; median = (15+30)/2 = 22.5 minutes. Use 22.5 as an imputed estimate only, not as a recovered measurement. Retain an imputation indicator and disclose the uncertainty.

Q24 — Checking preprocessing before bikes leave

(a) Median of all observed counts 10, 30, 50, 90 is (30+50)/2 = 40. Median fitted only on training counts 10, 30, 50 is 30. Fit the imputer using training data only and apply the same fit to held-out data; using all rows leaks test-set distribution information.

(b) An arbitrary numerical station code creates an unsupported order and distances between nominal stations. Use one-hot indicators (or another justified nominal representation) so a larger code is not interpreted as greater station size, risk or physical distance.

(c) Fault confirmation recorded after dispatch is unavailable at prediction time and leaks future information. Cycle count or station, stated as already available at dispatch, is usable. The timing of feature availability is essential.

Q25 — What separate data lists can tell you

(a) Positive pairing: (10,20), (20,40), (30,60), (40,80), giving increasing repair minutes as rentals rise. Negative pairing: (10,80), (20,60), (30,40), (40,20), giving decreasing repair minutes as rentals rise.

(b) Both pairings use the same individual rental and repair values, so they have the same marginal means and histograms, yet opposite associations. Daily/date-level pairings were lost. Recover the matching dates or original paired rows to assess association.

(c) A positive observational association is not proof that rentals cause repair time. Fleet size, bike age, weather, or inspection effort can affect both. Collect the potential confounder and compare matched or similar groups; even such comparisons do not guarantee causality.

Q26 — Compare stations before and after combining groups

(a) Low-use fault percentages: East = 2/20 = 10%, West = 6/40 = 15%. High-use: East = 18/30 = 60%, West = 7/10 = 70%. East is lower within both use groups.

(b) Pooled East = (2+18)/(20+30) = 20/50 = 40%. Pooled West = (6+7)/(40+10) = 13/50 = 26%. East has 60% high-use bikes versus West at 20%; high-use bikes have much higher fault rates. Different group mixtures reverse the aggregate ranking (Simpson’s paradox).

(c) With an equal low-/high-use mix, standardized East = 0.5×10%+0.5×60%=35%; West = 0.5×15%+0.5×70%=42.5%. East would then have the lower percentage. The observational records do not prove a causal station effect; age/condition and other factors may differ.

This sheet explains the expected methods. Equivalent correct reasoning and valid alternative methods are possible.

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